SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING
COURSE CODE: CSA43
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COURSE OUTCOME COVERED

CO-1: Develop well-structured, standards-compliant web pages using HTML/XHTML and XML, and apply Internet protocols and HTTP communication mechanisms for web content delivery. **(L2)**

Assessment Tool: Code along exercise: Make your first HTML page together with CSS
Weightage: 10%

QUESTIONS: Total Marks: 10

Scenario: Web Development Standards Implementation

A start up is developing a modern web application. During development, the team encounters multiple issues related to HTML elements, HTTP methods, XML syntax, and web communication protocols.
The following observations are made:

- The team needs to embed a promotional video on the homepage.
- A registration form must securely send user data to the server.
- The navigation menu must follow semantic HTML standards.
- XML configuration files are being used but contain syntax errors.
- The application must communicate over the correct web protocol.

Data Snippets Provided

Snippet 1: Video Embedding Options

```
A. <media src="video.mp4"></media>
B. <video src="video.mp4" controls></video>
C. <movie src="video.mp4"></movie>
D. <vid src="video.mp4"></vid>
```

Snippet 2: Form Submission Methods

```
A. GET
B. POST
C. PUT
D. DELETE
```

Snippet 3: Navigation Elements

```
A. <div>
B. <nav>
C. <section>
D. <header>
```

Snippet 4: XML Syntax

```
A. <note><to>User</to><from>Admin</from>
B. <note><to>User</to><from>Admin</from></note>
C. <note><to>User</from></note>
D. <note to="User"></note
```

Snippet 5: Web Communication Protocols

```
A. FTP
B. HTTP
C. SMTP
D. SNMP
```

Questions:

1. Select the **correct HTML tag for video embedding** and justify your answer. 2. Identify the **appropriate HTTP method for form submission** and explain why. 3. Choose the **proper semantic element for navigation** and justify its usage. 4. Identify the **correct XML syntax** and explain the error in other options. 5. Choose the **correct protocol for web communication** and justify your choice.

Code of Conduct:

*I **mrudhula.k(192372290)** (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.*

Signature of the student: mrudhula.k
RUBRICS FOR CALCULATION:

Criteria	**Weightage**	**Level 4 – Exemplary**	**Level 3 – Proficient**	**Level 2 – Developing**	**Level 1 – Beginning**
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding ; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts

Analysis 20	In-depth analysis with strong reasoning and insights Good analysis with logical reasoning Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design 20	Innovative, feasible solutions with strong justification Logical solutions with adequate justification Basic solutions with weak justification	Incorrect or no solution provided
Clarity 10	Clear, well structured, and easy to understand Organized and understandable presentation Limited clarity and structure	Poor clarity; difficult to understand

1. Select the Correct HTML Tag for Video

Embedding and Justify Your Answer

Correct Answer: B. <video src="video.mp4" controls></video>

Justification:

● The <video> tag is the standard HTML5 element used to embed videos in a web page. ● The controls attribute displays play, pause, volume, and fullscreen controls. ● It is supported by all modern web browsers.

Example:

```
<video src="video.mp4" controls></video>
```

2. Identify the Appropriate HTTP Method for Form Submission and Explain Why

Correct Answer: B. POST

Explanation:

- The **POST** method sends form data in the HTTP request body.
- It is more secure than **GET** because sensitive information is not displayed in the URL.
- It is commonly used for registration forms, login forms, and data submission.

Example:

```
<form action="/register" method="POST">
```

3. Choose the Proper Semantic Element for Navigation and Justify Its Usage

Correct Answer: B. <nav>

Justification:

- The <nav> element defines the navigation section of a webpage.
- It contains links to other pages or sections of the website.
- It improves accessibility, SEO, and follows HTML5 semantic standards.

Example:

```
<nav>
  <a href="home.html">Home</a>
  <a href="about.html">About</a>
  <a href="contact.html">Contact</a>
</nav>
```

4. Identify the Correct XML Syntax and Explain the Error in Other Options

Correct Answer: B.

```
<note>
  <to>User</to>
  <from>Admin</from>
</note>
```

Explanation of Other Options

Option A

```
<note><to>User</to><from>Admin</from>
```

Error: Missing closing `</note>` tag.

Option C

```
<note><to>User</from></note>
```

Error: Opening and closing tags do not match (`<to>` is closed with `</from>`).

Option D

```
<note to="User"></note
```

Error:

- Missing closing `>` in the `</note>` tag.
- Invalid XML syntax.

5. Choose the Correct Protocol for Web

Communication and Justify Your Choice

Correct Answer: B. HTTP

Justification:

- **HTTP (HyperText Transfer Protocol)** is the standard protocol used for communication between web browsers and web servers.
- It transfers web pages, images, videos, and other web resources.
- Secure websites use **HTTPS**, which is the encrypted version of HTTP.

Other Options

- **FTP** – Used for file transfer.
- **SMTP** – Used for sending emails.
- **SNMP** – Used for network management and monitoring.

Conclusion

The startup should use the HTML5 `<video>` tag for embedding videos, the **POST** method for secure form submission, the semantic `<nav>` element for navigation, well-formed XML syntax with matching opening and closing tags, and the **HTTP/HTTPS** protocol for browser-server communication. These practices ensure standards compliance, security, accessibility, and reliable web application development.